Absolute error

1

a 0.5 kg **b** 0.5 g **c** $5

2 $\frac{1}{2}$ mL

3

a 0.05 mL **b** 0.5 mm **c** 0.05 g **d** 0.5 s

e 0.005 kg **f** 0.05 m **g** 0.005 °C

4 We do not know what place value the number has been rounded to.

The limits of accuracy

5

a 1 cm **b** 90 cm

c 0.5 cm **d** **i** No **ii** No **iii** No **iv** Yes

6

a 10 kg **b** 5 kg

c Lower bound: 65 kg, upper bound: 75 kg

7

a 50 km/h **b** 5 km/h **c** 55 km/h

8

a **i** 13.455 km **ii** 13.445 km **b** **i** 6.45 km **ii** 6.35 km

c **i** 5.5 m **ii** 4.5 m **d** **i** 14.005 kg **ii** 13.995 kg

e **i** 4884.5 m **ii** 4883.5 m **f** **i** $52.50 **ii** $47.50

9 19.995 m

10 2.05 m

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a Maximum: 175, minimum: 165

b Maximum: 3050, minimum: 2950

12

a 26.25

b 0.6

13

a 13.85 m/s

b 13.95 m/s

Percentage error

14

a $\pm 0.91\%$

b $\pm 0.96\%$

c $\pm 7.14\%$

d $\pm 1.22\%$

e $\pm 0.29\%$

15

a

i 4.7 kg

ii 15.5%

b

i 0.07 m

ii 5.3%

c

i 7.2 minutes

ii 13.8%

16

a 15 891

b 16 109

c 18.9%

d 19.2%

e Italy's

Area and volume calculations

17

a $153.9\ \text{cm}^2$

b $132.7\ \text{cm}^2, 176.7\ \text{cm}^2$

c $22.8\ \text{cm}^2$

d 14.8%

18

a Upper bound: $272.6925\ \text{m}^2$, lower bound: $269.3925\ \text{m}^2$

b Upper bound: 66.2 m, lower bound: 65.8 m

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a Lower bound: $9.5\ \text{cm}$, upper bound: $10.5\ \text{cm}$

b $100\ \text{cm}^2$

c Lower bound: $90.25\ \text{cm}^2$, upper bound: $110.25\ \text{cm}^2$

d $1000\ \text{cm}^3$

e Lower bound: $857.375\ \text{cm}^3$, upper bound: $1157.625\ \text{cm}^3$

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